

Ankit Kumar

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Profile

CS student specializing in Java, Spring Boot, and scalable backend systems. Focused on database performance optimization and solving complex problems through Data Structures and Algorithms.

Skills

Java, Spring Boot, PostgreSQL, REST APIs, Redis, JavaScript/TypeScript, Next.js, Git

Experience

Backend Engineer Intern, ControlShift Talent Pvt. Ltd.

02/2025 – 05/2025

- Implemented **WebSocket architecture** for low-latency, real-time data features.
- Optimized **PostgreSQL indexing**, reducing API response latency by **40%**.
- Built an **async job system** using BullMQ & Redis, increasing throughput by **3x**.
- Engineered robust **REST APIs** with centralized validation and error handling.

Projects

CinemaCore, Java, Spring Boot, PostgreSQL, Stripe · github.com/ankitkr-04/cinematicore

- Built cinema booking system achieving zero double-bookings via pessimistic locking & ACID transactions
- Engineered seat reservation with SELECT FOR UPDATE locking → handles 50+ concurrent bookings safely
- Integrated Stripe webhooks with idempotency checks → automated payment confirmation & refunds
- Implemented hold timer with async jobs → 5-min auto-release, email notifications

RedisCore, Java, NIO, Concurrency · github.com/ankitkr-04/redis-core

- Developed a high-performance in-memory store using **Java NIO** to handle multiple concurrent client connections.
- Implemented **RESP protocol** compatibility, supporting core data types and atomic operations.
- Engineered **Master-Replica architecture** to handle data synchronization and improve system reliability.
- Optimized data retrieval via **thread-safe structures**, focusing on reducing lock contention and latency.

MuseoBot, Next.js, Clerk, PostgreSQL · github.com/ankitkr-04/museobot

- Integrated **Razorpay** with idempotency checks to prevent duplicate payment transactions.
- Developed an **Admin Dashboard** for real-time booking analytics and user activity tracking.
- Integrated **Botpress AI** to automate customer support and simulated booking workflows.
- Managed secure user authentication and session management using **Clerk**.

Education

Bachelor of Technology in Computer Science,
Lakshmi Narain College of Technology, Bhopal-462021
CGPA: 8.25

2023 – Present

Certificates

Oracle Cloud Infrastructure 2025 Developer Professional (1Z0-1084-25) – Oracle
Cloud Foundations – AWS Academy